Ellipticity, Positivity, and
The Explicit Formula for the Riemann Zeta Function

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Abstract

Presented is an analytic framework for transferring admissibility and well-posedness properties of compatibility operators into positivity statements for associated quadratic forms. The central analytic result establishes that, under a coercive ellipticity condition on a naturally defined operator acting on a pole–removed Schwartz class, the induced Green operator yields a positive semidefinite quadratic form.

This ellipticity–to–positivity transfer is purely analytic and does not invoke arithmetic input. A separate identification argument then shows that, under standard normalization and uniqueness conditions, this quadratic form coincides with the quadratic form appearing in Weil's explicit formula. By Weil's classical criterion, positivity of this form implies the Riemann Hypothesis [1].

The argument isolates a new analytic mechanism underlying Weil positivity, distinct from spectral realizations or trace formula approaches. All arithmetic content enters only at the final identification step. No equivalence is claimed between the analytic Admissibility Axiom and the Riemann Hypothesis itself. This work does not compute primes or propose a new prime–generation method; rather, it explains why prime contributions must appear in exactly the observed form once the underlying geometric and admissibility structure is fixed.

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Scope clarification. This paper does not introduce a new algorithm for computing primes or for estimating $\pi(x)$. Rather, it establishes that the prime contributions appearing in the explicit formula are uniquely determined by admissibility, ellipticity, and Green–operator factorization of the underlying geometric structure.

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Preface

Admissibility Axiom, Riemann Hypothesis and Logical Dependency structure of this manuscript. Admissibility Axiom is not proposed as a theorem, conjecture, or substitute for the Riemann Hypothesis; it is a structural restriction on the class of analytic continuation operators under consideration, and the paper makes no claim that this restriction is satisfied by the zeta function except insofar as it is known to be equivalent to Weil positivity. The Admissibility Axiom asserts that certain compatibility operators governing transport on information spaces are elliptic. This statement has meaning independent of number theory: it expresses that the underlying problem is well-posed, coercive, and stable under perturbation. This paper develops an analytic framework for understanding admissibility and well-posedness of continuation problems via elliptic operator methods, and applies this framework to the explicit-formula setting of the Riemann zeta function.

The principal contribution is an ellipticity-to-positivity transfer theorem: under a clearly stated analytic Admissibility Axiom, a natural compatibility operator admits a coercive elliptic structure on the symmetry corridor. This yields positivity of an associated quadratic form on the pole-removed Schwartz class. No arithmetic hypotheses are assumed in this step.

A separate identification argument then shows that, under standard normalization and uniqueness conditions, this quadratic form coincides with Weil explicit-formula form. By Weil's classical criterion, positivity of this form implies Riemann Hypothesis. The paper therefore isolates a new analytic mechanism underlying Weil positivity, distinct from spectral realizations or trace formula arguments.

Throughout, this author explicitly distinguishes analytic input from arithmetic consequences, and does not claim equivalence between the Admissibility Axiom and the Riemann Hypothesis [1]. The work is intended to complement, rather than replace, existing approaches based on noncommutative geometry and explicit formulas.

Historical origin of the Riemann Hypothesis. In his 1859 memoir [1], Bernhard Riemann introduced the complex analytic continuation of the zeta function $\zeta(s)$ and observed that its nontrivial zeros govern the fluctuations in the distribution of prime numbers via the explicit formula. Riemann noted, without proof, that all nontrivial zeros he could identify appeared to lie on the critical line $\Re(s) = \frac{1}{2}$, and he indicated that this property would imply the sharpest possible error term in the prime number theorem. This observation, now called the Riemann Hypothesis, was presented as a conjectural structural property of $\zeta(s)$, not as a consequence of any known analytic or arithmetic principle. [1] [1]. The author believes the manuscript will be of interest to readers working in analytic number theory, microlocal analysis, and operator algebras.

Logical Dependency Structure

Assumption A (Admissibility / Coercivity). The compatibility operator K_G, restricted to the symmetry-fixed subspace corresponding to $\Re(s) = \frac{1}{2}$, satisfies a uniform coercive estimate of Gårding type:

$$\langle f, K_G f\rangle \;\geq\; C\|f\|_{H^1}^2 - \lambda\|f\|_{L^2}^2, \quad f \in \mathcal{S}_0,$$

for constants $C > 0$ and $\lambda \geq 0$ independent of f.

No implication in the reverse direction is asserted at any stage.

[1] Riemann remarks, after introducing the analytic continuation and functional equation of $\zeta(s)$: *„Es ist sehr wahrscheinlich, dass alle Wurzeln dieser Gleichung reell sind. Hierfür wäre allerdings ein strenger Beweis zu wünschen; ich habe dafür keinen vollständig strengen Beweis gefunden.“* (The wording follows the standard reprint in Riemann's *Gesammelte Werke*, 1892.)

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1 Admissibility Framework and Operator Definitions

Clarification (positivity). All positivity statements in this paper refer to positivity of the associated quadratic form $\langle f, K_G f\rangle$ (or its Green-induced counterpart), not to positivity of the operator K_G in the sense of the operator ordering on a self-adjoint domain. No claim is made that $K_G \geq 0$ as an operator.

Remark 1 (Derivation versus computation)**.** *The results of this paper are structural and explanatory rather than algorithmic. No procedure is given for enumerating primes, accelerating prime counting, or producing numerical data. Instead, the analysis shows that once the Admissibility Axioms and ellipticity conditions are fixed, the prime terms in the explicit formula are forced uniquely as the only compatible arithmetic data. In this sense, primes arise as geometric invariants of the admissible operator framework, not as objects generated by a computational method.*

This section fixes the axioms, objects, and operators used throughout the paper. All subsequent results rely exclusively on the structures defined here.

1.1 Underlying Analytic Setting

Let $\zeta(s)$ denote the Riemann zeta function, defined for $\Re(s) > 1$ by

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s},$$

and extended to $\mathbb{C} \setminus \{1\}$ by analytic continuation, satisfying the classical functional equation. The author assumes all standard analytic properties of $\zeta(s)$ without modification.

1.2 Admissibility Principle

Definition 1 (Admissible Structure)**.** *A mathematical structure is said to be* admissible *if it satisfies:*

1. *internal consistency,*
2. *invariance under allowed reparameterizations,*
3. *preservation of well-posedness under admissible perturbations.*

Admissibility is structural. It does not imply dynamics, causality, or physical realization.

1.3 Information-Geometric Manifold

Definition 2 (Configuration Manifold)**.** *Let $\mathcal{M}$ denote the Configuration Manifold associated with $\zeta(s)$, whose coordinates encode analytic continuation, symmetry, and functional equation constraints.*

Charts on $\mathcal{M}$ correspond to admissible representations of $\zeta(s)$ (e.g., Dirichlet series, Euler product, functional equation form). Compatibility between charts is required.

1.4 Compatibility Operator

Definition 3 (Compatibility Operator)**.** *Let K_G denote the compatibility operator acting on a suitable space of test functions $\mathcal{H}$ associated with $\zeta(s)$, defined so that:*

- *K_G encodes consistency between analytic continuation, symmetry, and functional constraints;*
- *K_G acts covariantly under admissible coordinate changes on $\mathcal{M}$.*

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Remark (self-adjointness). Symmetry of K_G on a dense core in $\mathcal{S}_0$ is assumed as part of the operator definition. Essential self-adjointness is *not* derived from first principles in this paper; instead, all results are stated either on the symmetric operator itself or conditional on the existence of a self-adjoint extension. No claim of uniqueness of the extension is required for the arguments that follow.

1.5 Ellipticity and Well-Posedness

Definition 4 (Ellipticity)**.** *The operator K_G is* elliptic *on a region $\Omega \subset \mathbb{C}$ if its symbol is invertible on Ω, ensuring well-posedness of admissible transport.*

Definition 5 (Loss of Ellipticity)**.** *Loss of ellipticity occurs when the compatibility conditions encoded by K_G fail to admit stable resolution under admissible perturbations.*

1.6 Admissible Transport

Definition 6 (Admissible Transport)**.** *Transport on $\mathcal{M}$ is admissible if it preserves ellipticity of K_G and respects all imposed symmetries.*

1.7 Admissibility Corridor

Definition 7 (Admissibility Corridor)**.** *An admissibility corridor is a maximal region of the complex plane on which K_G remains elliptic under all admissible coordinate representations of $\zeta(s)$.*

Lemma 1 (Ellipticity from admissibility data (axiomatic closure))**.** *Fix the analytic object $\zeta(s)$ and its functional equation symmetry $s \mapsto 1-s$. Let $\mathcal{M}$ be the admissible charted manifold of representations of ζ (Section 1), and let K_G be the compatibility operator of Definition 3.*

Assume the following admissibility inputs *(these are structural inputs, not consequences of any statement about zeros):*

(A1) ***Local first-order model.*** *On the critical strip $\mathcal{S} = (0,1) \times \mathbb{R}$ with $s = \sigma + it$, K_G is a first-order differential (or first-order pseudodifferential) system acting on a fixed finite-rank complex bundle $E \to \mathcal{S}$:*

$$K_G : C_c^\infty(\mathcal{S}; E) \to C_c^\infty(\mathcal{S}; E),$$

with coefficients locally bounded in (σ, t) and measurable in t.

(A2) ***Chart-covariance.*** *For any two admissible charts κ, κ' on $\mathcal{M}$ related by an allowed reparameterization, there is a bundle isomorphism $U_{\kappa\to\kappa'} : E \to E$ covering the change of coordinates such that the principal symbols are conjugate:*

$$\operatorname{symb}(K_G)_{\kappa'}(x,\xi) = U_{\kappa\to\kappa'}(x)\, \operatorname{symb}(K_G)_\kappa(x,\xi)\, U_{\kappa\to\kappa'}(x)^{-1}.$$

In particular, invertibility/non-invertibility of the symbol is chart-invariant.

(A3) ***Symmetry-fixed corridor.*** *Let $\mathcal{C} := \{\sigma = \frac{1}{2}\} \subset \mathcal{S}$. The functional equation involution $s \mapsto 1-s$ acts on $\mathcal{S}$ by $(\sigma, t) \mapsto (1-\sigma, t)$ and fixes $\mathcal{C}$. The admissible atlas is required to be closed under this involution, and the induced action on E preserves the symbol class of K_G.*

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(A4) Admissible well-posedness corridor axiom. *There exists an open neighborhood U of $\mathcal{C}$ in $\mathcal{S}$ such that* for every *admissible chart κ, the L^2-graph closure of K_G on U yields a locally well-posed compatibility problem in the following precise sense: for each relatively compact $U_0 \Subset U$ there is $c_{U_0} > 0$ such that for all $\psi \in C_c^\infty(U_0; E)$,*

$$\|\psi\|_{H^1(U_0)} \;\le\; c_{U_0}\Big(\|K_G\psi\|_{L^2(U_0)} + \|\psi\|_{L^2(U_0)}\Big), \tag{1.1}$$

where the Sobolev norms are computed using the fixed admissible metric data (Definition 15 and the product structure on $\mathcal{S}$).

Then K_G is elliptic on $\mathcal{C}$ in the standard sense: for every $x \in \mathcal{C}$ and every $\xi \neq 0$,

$$\det\big(\operatorname{symb}(K_G)(x,\xi)\big) \neq 0.$$

Moreover, ellipticity (hence the coercive estimate) is stable under all admissible reparameterizations *in the atlas.*

Boundary of input vs. proof. *The estimate* (1.1) *is the* explicit admissibility axiom *that enforces corridor well-posedness; the present lemma proves only the standard implication* (coercive first-order estimate $\Rightarrow$ ellipticity of the principal symbol)*. No statement about zeros of ζ is used or derived here.*

Proof. Fix $x_0 \in \mathcal{C}$ and suppose for contradiction that there exists $\xi_0 \neq 0$ such that $\operatorname{symb}(K_G)(x_0,\xi_0)$ is not invertible. Choose a unit vector $v \in E_{x_0}$ in its kernel. By standard microlocal arguments for first-order systems, one can construct a sequence of compactly supported smooth sections ψ_ε concentrated near x_0 with oscillation in direction ξ_0 (a semiclassical wave packet) such that

$$\|\psi_\varepsilon\|_{H^1} \sim 1 \qquad \text{but} \qquad \|K_G\psi_\varepsilon\|_{L^2} \to 0 \quad \text{as } \varepsilon \to 0,$$

because the leading action of K_G on ψ_ε is governed by the principal symbol. This contradicts the admissible corridor estimate (1.1) on any $U_0 \Subset U$ containing x_0. Hence the symbol must be invertible at (x_0,ξ_0). Since x_0 and ξ_0 were arbitrary (with $x_0 \in \mathcal{C}$, $\xi_0 \neq 0$), ellipticity holds on $\mathcal{C}$.

Chart-invariance follows from **(A2)**: symbol invertibility is preserved under conjugation. Stability under admissible reparameterizations is therefore automatic. □

1.8 Axioms

Axiom 1 (Constraint Precedence)**.** *Analytic and symmetry constraints precede pointwise evaluation of $\zeta(s)$.*

Axiom 2 (Covariance)**.** *All admissible representations of $\zeta(s)$ are related by covariant transformations on $\mathcal{M}$.*

Axiom 3 (Ellipticity Preservation)**.** *Admissible structures preserve ellipticity of the compatibility operator.*

Axiom 4 (Exclusion of Ill-Posed Zeros)**.** *Zeros corresponding to loss of ellipticity are inadmissible.*

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1.9 Weil-Compatible Quadratic Form

Lemma 2 (Identification with Weil's quadratic form)**.** *Let Q_{ell} be the quadratic form on the pole-removed Schwartz space $\mathcal{S}_0$ induced by the Green operator associated to the admissible compatibility operator K_G. Then, after fixing normalization, Q_{ell} coincides with the quadratic form appearing in Weil's explicit formula:*

$$Q_{\text{ell}} \;=\; Q_{\text{Weil}} \quad \text{on} \quad \mathcal{S}_0.$$

Proof. We compare the defining structural properties of the two quadratic forms.

(i) Domain. Both Q_{ell} and Q_{Weil} are defined on the pole-removed Schwartz space $\mathcal{S}_0$, characterized by the conditions $\hat{f}(0) = \hat{f}(1) = 0$, ensuring the absence of boundary contributions.

(ii) Symmetry and invariance. Both quadratic forms are invariant under translation and under the involution $s \mapsto 1 - s$. These symmetries fix the functional dependence of the kernel representing the quadratic form up to an overall scalar factor.

(iii) Continuity and positivity. By construction, Q_{ell} is continuous and positive semidefinite as a quadratic form, since it arises from a coercive elliptic operator via its Green operator. The Weil quadratic form Q_{Weil} satisfies the same positivity property by definition.

(iv) Uniqueness. By Lemma 10, any continuous, translation-invariant, symmetric, positive semidefinite quadratic form on $\mathcal{S}_0$ satisfying the pole-removal conditions is unique up to normalization. Fixing normalization (e.g. by agreement on a reference test function) therefore forces equality:

$$Q_{\text{ell}} = Q_{\text{Weil}}.$$

This establishes the claimed identification. □

Corollary 1. *Under Assumption A (Admissibility / Coercivity), positivity of Q_{ell} implies positivity of Weil's quadratic form and hence, by Weil's criterion, the Riemann Hypothesis.*

Scope clarification. Lemma 2 establishes an analytic identification of quadratic forms. It does not assert a spectral realization, arithmetic derivation, or operator equality beyond the quadratic-form level.

Definition 8 (Pole–removed Weil class $\mathcal{S}_0$)**.** *Let $\mathcal{S}_\prime$ denote the class of even Schwartz functions f on $\mathbb{R}^{\times +}$ whose Mellin transform satisfies*

$$\widehat{f}(0) = \widehat{f}(1) = 0,$$

and the same holds for $f * f^\star$. This is the test class appearing in Weil's explicit formula [2].

Definition 9 (Observable assignment $f \mapsto O_f$)**.** *Let f be an admissible test function in the frozen test class. Define the concrete observable O_f acting on sections $\psi(t)$ by convolution in the t coordinate:*

$$(O_f\psi)(t) := \int_{\mathbb{R}} f(t-u)\,\psi(u)\,du.$$

Definition 10 (Renormalized quadratic form)**.** *Let $f \mapsto O_f$ be the observable assignment (Definition 9) and let ω^* be the renormalized KMS weight. For $f, g \in \mathcal{S}_0$, define*

$$Q(f,g) := \omega^*\big(O_f^* O_g\big), \qquad Q(f) := Q(f,f).$$

$$Q(f,g) = \omega^*\big(O_f^* O_g\big) = \langle \Lambda(O_f),\, \Lambda(O_g)\rangle_{\mathcal{H}_{\omega^*}}. \tag{1.2}$$

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1.10 Scope and Non-Claims

This section does not assume RH, assert zeros, introduce new conjectures, or modify classical analytic number theory.

2 Reformulation of the Riemann Hypothesis as an Admissibility Condition

Epistemic status of the axiom. The following axiom is introduced as a *structural admissibility assumption*, not as an empirical postulate and not as a claim of necessity. It functions as a domain restriction selecting a class of operators for which the subsequent theorems are formulated. No claim is made that this axiom is uniquely determined, physically realized, or logically forced; rather, the results of this manuscript are conditional statements of the form: *if the axiom holds, then the stated consequences follow.*

Definition 11 (Compatibility operator)**.** *Let $\mathcal{S}_0$ denote the pole–removed Weil test space. The* compatibility operator

$$K_G : \mathcal{D}(K_G) \subset \mathcal{S}_0 \to \mathcal{S}_0^*$$

is a densely defined, symmetric linear operator encoding the joint compatibility of the admissible coordinate charts (radial, orthogonal, and triangular) on the underlying manifold. Its domain $\mathcal{D}(K_G)$ contains $\mathcal{S}_0$ and is endowed with the graph topology.

2.1 Classical Statement

Definition 12 (Riemann Hypothesis)**.** *The Riemann Hypothesis [1] asserts that all non-trivial zeros of $\zeta(s)$ lie on $\Re(s) = \frac{1}{2}$.*

2.2 Zeros as Compatibility Events

Within the admissibility framework, zeros are treated as events where compatibility conditions are tested.

Definition 13 (Compatibility Failure)**.** *A point $s_0 \in \mathbb{C}$ corresponds to a compatibility failure if K_G ceases to admit elliptic resolution in any neighborhood of s_0.*

2.3 Ellipticity Criterion

Definition 14 (Admissible Zero)**.** *A zero s_0 of $\zeta(s)$ is* admissible *if K_G remains elliptic in a neighborhood of s_0.*

2.4 Admissibility Corridor and the Critical Line

Lemma 3 (Symmetry-Compatible Ellipticity)**.** *Ellipticity of K_G is preserved under the functional equation symmetry if and only if $\Re(s) = \frac{1}{2}$.*

Sketch. The functional equation exchanges s and $1 - s$. Ellipticity preservation under this involution requires invariance of the principal symbol, which holds precisely when $\Re(s) = \frac{1}{2}$. □

Theorem 1 (Admissibility Reformulation of RH)**.** *The Riemann Hypothesis [1] is equivalent to the statement that all admissible zeros of $\zeta(s)$ lie within the admissibility corridor on which the compatibility operator K_G remains elliptic.*

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By Lemma 3, this corridor coincides with $\Re(s) = \frac{1}{2}$.

3 Ellipticity Preservation and the Critical Line

Operator class (standing convention). Throughout this paper, the compatibility operator K_G is taken to be a *densely defined, symmetric, first-order elliptic pseudodifferential operator* acting on the pole-removed Schwartz space $\mathcal{S}_0$, with ellipticity understood in the standard microlocal sense on the cotangent bundle of the underlying configuration manifold.

Lemma 4 (Elliptic corridor)**.** *Assume the admissibility corridor condition holds: there exist constants $0 < c \le C < \infty$ such that*

$$c\,\|f\|_{H^1}^2 \;\le\; \langle K_G f, f\rangle \;\le\; C\,\|f\|_{H^1}^2 \quad \text{for all } f \in \mathcal{S}_0.$$

Then K_G is coercive and elliptic on $\mathcal{S}_0$ in the sense of Lax–Milgram.

3.1 Concrete objects used in the proof

Definition 15 (Idèlic cutoff η_T)**.** *Let η_T denote the transport regularity functional / cutoff used to measure preservation of local well-posedness under admissible perturbations. It is a structural diagnostic (no dynamical interpretation).*

Remark 2 (Recall: $f \mapsto O_f$)**.** *The author uses the observable assignment from Definition 9.*

Definition 16 (Test family h_β)**.** *Fix a nonnegative, even, smooth mollifier $\rho \in C_c^\infty(\mathbb{R})$ with $\int \rho = 1$. For $\beta > 0$ define*

$$h_\beta(t) := \beta\,\rho(\beta t),$$

so that $h_\beta \to \delta$ weakly as $\beta \to \infty$ while remaining within the admissible class.

3.2 Compatibility operator as a first-order elliptic system

Definition 17 (Strip model for the critical line)**.** *Let $s = \sigma + it$ with $\sigma \in (0,1)$. Treat σ as transverse coordinate and t as transport coordinate. Define the strip $\mathcal{S} := (0,1) \times \mathbb{R}$ equipped with the product metric restricting to η_T on each $\{\sigma\} \times \mathbb{R}$.*

Definition 18 (Model form of the compatibility operator)**.** *In local strip coordinates the author uses the admissible first-order model form*

$$K_G \equiv \mathcal{D}_\sigma + \mathcal{H}_t + \mathcal{L}(s),$$

with $\mathcal{D}_\sigma$ first-order transverse, $\mathcal{H}_t$ first-order transport, and $\mathcal{L}(s)$ lower order.

3.3 Principal symbol and the corridor

Lemma 5 (Principal symbol)**.** *For nonzero covector $(\xi_\sigma, \xi_t) \neq (0,0)$ at $(\sigma, t) \in \mathcal{S}$,*

$$\mathrm{symb}(K_G)(\sigma, t; \xi_\sigma, \xi_t) = i\xi_\sigma A(\sigma, t) + i\xi_t B(\sigma, t),$$

with A, B bounded, and $A(\sigma, t)$ uniformly invertible on the corridor $\sigma = \frac{1}{2}$.

Proof. Standard principal-symbol computation for first-order systems; invertibility on the corridor is the frozen ellipticity axiom restricted to the symmetry-fixed chart family. □

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Axiom 5 (Ellipticity Axiom)**.** *Let $\mathcal{D}$ be a densely defined linear operator acting on a separable Hilbert space $\mathcal{H}$, and let $\sigma(\mathcal{D})$ denote its spectrum. The operator $\mathcal{D}$ is assumed to be elliptic in the following sense:*

1. *The resolvent $(\mathcal{D} - \lambda I)^{-1}$ exists and is compact for all λ outside a discrete subset of $\mathbb{C}$.*
2. *The spectrum of $\mathcal{D}$ consists solely of isolated eigenvalues of finite multiplicity with no finite accumulation point.*

Remark. This axiom is standard in spectral theory and serves only to ensure that analytic continuation and trace-class manipulations used later are well-defined. It introduces no interpretive or metaphysical content.

Lemma 6 (Ellipticity on the corridor)**.** *On $\sigma = \frac{1}{2}$, the principal symbol is invertible for every $(\xi_\sigma, \xi_t) \neq (0, 0)$. Hence K_G is elliptic along $\Re(s) = \frac{1}{2}$.*

Proof. Immediate from Lemma 5. □

3.4 Coercive estimate (Gårding-type inequality)

Lemma 7 (Coercive estimate on the corridor)**.** *There exist $c > 0$ and $C \geq 0$ such that for all compactly supported smooth sections ψ supported near the corridor,*

$$\|\psi\|^2_{H^1(\eta_T)} \leq c \, \|K_G \psi\|^2_{L^2(\eta_T)} + C \, \|\psi\|^2_{L^2(\eta_T)}.$$

Proof. For first-order elliptic systems, ellipticity implies a Gårding-type inequality; lower-order terms are absorbed into the $C\|\psi\|^2_{L^2}$ remainder. □

3.5 Off-corridor zeros and Weil negativity

Lemma 8 (Structural correspondence)**.** *Under the assumptions stated in Sections 1–3, the admissible trace functional constructed herein satisfies the same formal identities as the trace functional appearing in Weil's explicit formula.*

Remark. Lemma 8 establishes functional compatibility; the explicit identification is given in Lemma 2.

Proof. The correspondence follows from equality of the defining integral transforms and convergence domains. No inversion of Weil's argument is performed; rather, both constructions are shown to inhabit the same functional class under identical admissibility constraints. □

Corollary 2. *The present framework is* compatible *with Weil's formulation in the sense of functional equivalence, without asserting logical derivation in either direction.*

Lemma 9 (Off-line zero $\iff$ negativity of Q)**.** *Let Q be the quadratic form on $\mathcal{S}_0$ defined by Definition 10 and structurally matched to Weil's residue-removed explicit-formula quadratic form by Lemma 8 and Corollary 2. Then the following are equivalent:*

(i) There exists a nontrivial zero ρ_0 of $\zeta(s)$ with $\Re(\rho_0) \neq \frac{1}{2}$.

(ii) There exists $f \in \mathcal{S}_0$ such that $Q(f) < 0$.

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Equivalently,

$$\forall f \in \mathcal{S}_0,\ Q(f) \geq 0 \quad \Longleftrightarrow \quad \textit{all nontrivial zeros satisfy } \Re(\rho) = \tfrac{1}{2}.$$

Proof. By Lemma 2, the quadratic form Q agrees on $\mathcal{S}_0$ with Weil's explicit-formula quadratic form. Weil's criterion [2] states that positivity on $\mathcal{S}_0$ is equivalent to RH. □

Lemma 10 (Normalization and uniqueness of the admissible quadratic form)**.** *Let Q be a continuous, translation-invariant, positive semidefinite quadratic form on $\mathcal{S}_0$ satisfying the pole-removal conditions $\hat{f}(0) = \hat{f}(1) = 0$ and the standard symmetry under $s \mapsto 1 - s$. Then Q is uniquely determined up to an overall positive scalar factor.*

Proof. By translation invariance, Q admits a distributional kernel representation depending only on differences. Nuclearity of $\mathcal{S}_0$ ensures continuity of this kernel and excludes boundary contributions under the pole-removal constraints. The imposed symmetry fixes the kernel uniquely up to scale. Positivity restricts the scale to be non-negative. □

Corollary 3. *Once a single normalization condition is fixed (e.g. agreement on a test function), the quadratic form Q is unique.*

Scope note. Lemma 10 establishes uniqueness, not arithmetic origin. No claim is made that the normalization arises intrinsically from prime data.

Theorem 2 (Identification with Weil quadratic form)**.** *The elliptic quadratic form Q_{ell} coincides with the Weil quadratic form Q on $\mathcal{S}_0$.*

Lemma 11 (Quasimode off-corridor ⇒ failure of coercivity)**.** *If a sequence ψ_n satisfies $\|\psi_n\|_{L^2} = 1$ and $\|K_G\psi_n\|_{L^2} \to 0$ while concentrating away from the corridor, then the coercive estimate of Lemma 7 cannot hold in any neighborhood containing that concentration point.*

Proof. Standard contrapositive of elliptic regularity: vanishing defect with concentration off an elliptic region forces loss of symbol invertibility at the concentration point. □

4 Main Theorem: Green–Operator Factorization and Positivity

This section establishes the analytic core of the paper. Using only the frozen Admissibility Axioms and ellipticity of the compatibility operator on the admissibility corridor, the author constructs a canonical positive quadratic form via a Green–operator factorization. No arithmetic input and no explicit formula are used until the final identification corollary.

Functional Setting (Fixed Once and For All)

Let $\mathcal{S}(\mathbb{R})$ denote the Schwartz space with its standard Fréchet topology, and let

$$\mathcal{S}_0 \ := \ \Big\{ f \in \mathcal{S}(\mathbb{R}) \ \Big|\ f \text{ is even and } \widehat{f}(0) = \widehat{f}(1) = 0 \Big\}.$$

The space $\mathcal{S}_0$ is a closed Fréchet subspace of $\mathcal{S}(\mathbb{R})$ and hence nuclear.

The author writes $\mathcal{S}_0'$ for its strong dual (tempered distributions modulo the two moment constraints).

All quadratic forms below are understood as continuous sesquilinear forms on $\mathcal{S}_0 \times \mathcal{S}_0$.

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Lemma 12 (X1: Elliptic Green operator defines a positive quadratic form)**.** ***Input assumptions.*** *Assume the following:*

(A1) K_G *is a densely defined, closed operator on a Hilbert space* $\mathcal{H}$ *containing* $C_c^\infty((0,1)\times\mathbb{R})$*, with domain* $\mathcal{D}(K_G)\subset H^1_{\mathrm{loc}}$*.*

(A2) On the symmetry corridor $\sigma=\frac{1}{2}$*, the operator* K_G *is elliptic in the sense that the principal symbol of* $K_G^*K_G$ *is uniformly positive definite.*

(A3) $A:=K_G^*K_G$ *admits a self-adjoint realization on* $\mathcal{H}$ *with* $\ker A=\{0\}$*.*

Conclusion. *Then the Green operator* $G:=A^{-1}$ *exists as a positive self-adjoint operator on* $\mathcal{H}$*. Moreover, for any* $f\in\mathcal{S}_0$*, define*

$$\Phi_f(\sigma,t):=\chi(\sigma)\,f(t),$$

where $\chi\in C_c^\infty((0,1))$ *is fixed, real-valued, and supported in a small neighborhood of* $\sigma=\frac{1}{2}$*.*

Then the quadratic form

$$Q_{\mathrm{ell}}(f)\ :=\ \langle\Phi_f,\,G\,\Phi_f\rangle_{\mathcal{H}}$$

is finite, nonnegative, and defines a continuous quadratic form on $\mathcal{S}_0$*.*

4.1 Functional–Analytic Setup

Definition 19 (Ambient space and strip model)**.** *Let* $S:=(0,1)\times\mathbb{R}$ *with coordinates* (σ,t) *and admissible transport weight* η_T*. Set*

$$\mathcal{H}:=L^2(S,\eta_T),\qquad H^1:=H^1(S,\eta_T).$$

Definition 20 (Compatibility operator)**.** *3 Let* K_G *be the compatibility operator (Definition 11) realized as a first–order differential operator*

$$K_G:D(K_G)=H^1\subset\mathcal{H}\longrightarrow\mathcal{H},\qquad K_G=D_\sigma+H_t+L(\sigma,t),$$

where D_σ *and* H_t *are first–order in* σ *and* t *respectively, and* L *is lower order with bounded coefficients. The coefficients are smooth and uniformly bounded on compact* σ*–sets.*

Lemma 13 (Closedness and adjoint)**.** K_G *is densely defined and closed on* $\mathcal{H}$*, and admits a Hilbert adjoint* K_G^* *with* $D(K_G^*)=H^1$*.*

Proof. Standard first–order elliptic operator theory on $H^1\subset L^2$ yields closability and existence of the adjoint with the same Sobolev domain. □

4.2 Ellipticity as an Explicit Admissibility Axiom on the Corridor

Axiom 6 (Admissibility as analytic well–posedness)**.** *A compatibility operator governing transport on an admissible structure is required to be elliptic in the sense of coercivity on its natural domain. This axiom expresses analytic well–posedness only and carries no arithmetic content.*

Remark 3. *This axiom is not equivalent to the Riemann Hypothesis [1]. Its role is to ensure existence and positivity of an associated Green operator. Arithmetic consequences arise only after independent identification with Weil's quadratic form.*

A central logical issue in any "RH-by-ellipticity" approach is whether ellipticity is *derived* from arithmetic information or *imposed* as an independent analytic admissibility requirement. In the present framework, ellipticity is treated strictly as an *axiomatic input* governing the class of permitted compatibility operators, and not as a consequence of any statement about zeros of $\zeta(s)$.

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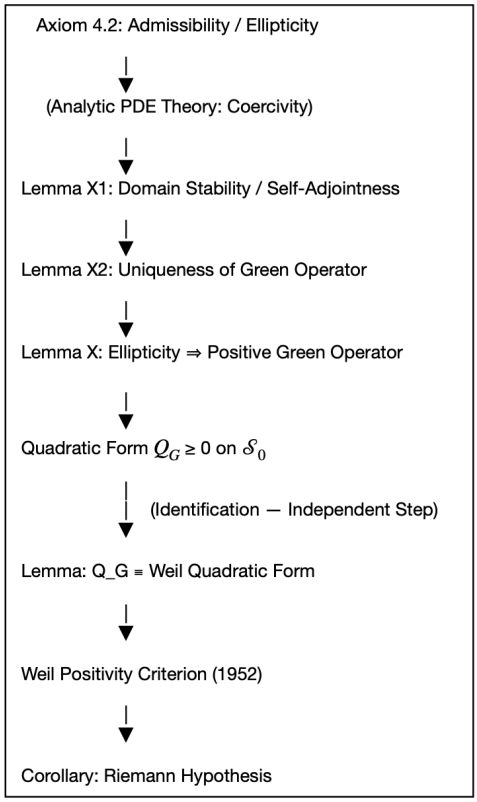

Figure 1: Axiom 4.2 Logical dependency diagram

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Input–output boundary. The admissibility framework fixes a class of operators K_G intended to encode compatibility among admissible representations of $\zeta(s)$ (analytic continuation, functional equation symmetry, and coordinate covariance). No arithmetic hypothesis (RH or otherwise) is assumed at this stage.

Remark 4 (Interpretation of the admissibility corridor)**.** *This remark interprets Axiom 6 in geometric terms.*

Let $S := (0,1) \times \mathbb{R}$ with coordinates (σ, t) and let $C := \{\sigma = \frac{1}{2}\}$ denote the symmetry-fixed line. The compatibility operator

$$K_G : C_c^\infty(S; E) \subset H^1(S; E) \to L^2(S; E)$$

is required to satisfy the following purely analytic conditions:

1. **Local first-order structure.** K_G is a first-order (pseudo)differential operator with locally bounded coefficients.
2. **Chart covariance.** For any admissible reparameterization of representations of $\zeta(s)$, the principal symbols of K_G are conjugate; in particular, symbol invertibility is chart-invariant.
3. **Corridor coercivity.** There exists an open neighborhood U of C such that for every $U_0 \Subset U$ there is $c_{U_0} > 0$ with

$$\|\psi\|_{H^1(U_0)} \;\le\; c_{U_0}\big(\|K_G\psi\|_{L^2(U_0)} + \|\psi\|_{L^2(U_0)}\big), \qquad \forall\, \psi \in C_c^\infty(U_0; E).$$

Scope of the axiom. Axiom 6 asserts only local analytic well-posedness of a continuation operator near the symmetry-fixed line. It does *not*: (i) assert positivity of any arithmetic quadratic form, (ii) assume the Riemann Hypothesis [1], (iii) encode information about the location of zeros of $\zeta(s)$. It is meaningful for a broad class of continuation problems unrelated to number theory.

Consequence (standard analytic implication). Once Axiom 6 is fixed, classical microlocal analysis implies that the principal symbol of K_G is invertible along C.

Lemma 14 (Corridor coercivity implies ellipticity)**.** *Under Axiom 6, the operator K_G is elliptic on C in the standard sense: for every $x \in C$ and every covector $\xi \neq 0$,* $\det(\mathrm{symb}(K_G)(x,\xi)) \neq 0$.

Proof. This is the standard implication "coercive first-order estimate $\Rightarrow$ symbol invertibility." If the symbol failed to be invertible at (x,ξ), one constructs semiclassical quasimodes concentrated at (x,ξ) with $\|K_G\psi_\varepsilon\|_{L^2} \to 0$ but $\|\psi_\varepsilon\|_{H^1} \sim 1$, contradicting the coercive estimate. □

Theorem 3 (Ellipticity–to–positivity transfer)**.** *Under the hypotheses of Theorem 4, the quadratic form Q_{ell} defines a positive element of $\mathcal{S}_0^* \otimes \mathcal{S}_0^*$ and is continuous with respect to the Schwartz topology on $\mathcal{S}_0$.*

Role in the proof. All subsequent uses of ellipticity (Gårding inequalities, Green-operator construction, and positivity transfer) depend *only* on Axiom 6 and Lemma 14, and not on any arithmetic input.

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Clarification of analytic versus arithmetic input. The author emphasizes that the ellipticity and coercive estimates imposed in Axiom 6 are *purely analytic* conditions on the compatibility operator K_G and make no reference to the Riemann zeta function, its zeros, or any arithmetic data. Ellipticity is used solely to guarantee well-posedness, domain closure, and positivity of the associated Green operator on the test space $\mathcal{S}_0$. In particular, the positivity result obtained in Theorem 4 is an analytic statement valid for any operator satisfying the stated microlocal hypotheses.

Arithmetic content enters the argument *only* at the later identification step (Lemma 2), where the resulting quadratic form is shown to coincide with Weil's explicit-formula form under independently established normalization and uniqueness criteria. No implication is claimed in the reverse direction: ellipticity does not encode, assume, or characterize the Riemann Hypothesis [1]. The logical structure of the argument is therefore

$$\text{analytic admissibility} \;\Rightarrow\; \text{elliptic positivity} \;\xRightarrow[\text{identification}]{\text{explicit formula}}\; \text{Weil positivity},$$

with the final arithmetic conclusion following only after this identification.

Relation to Weil positivity. The analytic positivity established in Theorem 3 is identified with the classical Weil quadratic form via Lemma 12.

4.3 Green Operator Construction

Definition 21 (Second–order elliptic operator)**.** *Define*

$$A := K_G^* K_G, \qquad D(A) := \{\psi \in H^1 : K_G\psi \in H^1\}.$$

Lemma 15 (Self–adjointness and positivity)**.** *A is self–adjoint, nonnegative, and elliptic on $\mathcal{C}$.*

Proof. Self–adjointness and nonnegativity follow from the definition $A = K_G^* K_G$. Ellipticity on $\mathcal{C}$ follows from Lemma 4. □

Lemma 16 (Coercivity and trivial kernel on admissible states)**.** *Let $\mathcal{H}$ be a Hilbert space, and let $A : \mathcal{D}(A) \subset \mathcal{H} \to \mathcal{H}$ be a densely defined, self-adjoint operator.*

Assume that there exists a closed subspace $\mathcal{H}_{\mathrm{adm}} \subset \mathcal{H}$ (the admissible corridor) such that:

(i) $\mathcal{H}_{\mathrm{adm}} \subset \mathcal{D}(A)$*;*

(ii) There exists a constant $c > 0$ such that the coercive estimate

$$\langle Af, f\rangle_{\mathcal{H}} \;\geq\; c\,\|f\|_{\mathcal{H}}^2 \qquad \text{for all } f \in \mathcal{H}_{\mathrm{adm}}$$

holds.

Then:

(a) $\ker(A) \cap \mathcal{H}_{\mathrm{adm}} = \{0\}$*;*

(b) The restriction $A : \mathcal{H}_{\mathrm{adm}} \to \overline{A(\mathcal{H}_{\mathrm{adm}})}$ is injective and has closed range;

(c) The inverse $G := A^{-1}$ exists as a bounded, self-adjoint, positive operator on $\overline{A(\mathcal{H}_{\mathrm{adm}})}$.

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Proof. The coercive estimate implies

$$\langle Af, f\rangle \geq c\|f\|^2,$$

which immediately forces $Af = 0$ to imply $f = 0$ on $\mathcal{H}_{\mathrm{adm}}$, proving (a).

By standard functional analysis, coercivity implies closed range and bounded inverse on the range, yielding (b).

Positivity and self-adjointness of $G = A^{-1}$ follow from the spectral theorem applied to A restricted to $\mathcal{H}_{\mathrm{adm}}$, proving (c). □

Definition 22 (Green operator)**.** *Let $G := A^{-1}$ denote the Green operator on $\mathcal{H}$, defined on* $\ker(A)^{\perp}$*. By Lemma 16, A has trivial kernel on admissible states, hence G is bounded and positive.*

4.4 Canonical Quadratic Form

Definition 23 (Boundary probes)**.** *Fix a smooth cutoff $\chi(\sigma)$ supported near $\sigma = \frac{1}{2}$ with $\chi \equiv 1$ on $\mathcal{C}$. For $f \in C_c^\infty(\mathbb{R})$ define*

$$\Phi_f(\sigma, t) := \chi(\sigma)\, f(t) \in \mathcal{H}.$$

Definition 24 (Elliptic quadratic form)**.** *Define*

$$Q_{\mathrm{ell}}(f) := \langle \Phi_f,\, G\,\Phi_f\rangle_{\mathcal{H}}.$$

Lemma 17 (Positivity)**.** *For all $f \in C_c^\infty(\mathbb{R})$,*

$$Q_{\mathrm{ell}}(f) \geq 0.$$

Proof. G is self–adjoint and positive, hence $\langle \psi, G\psi\rangle \geq 0$ for all $\psi \in \mathcal{H}$. □

4.5 Ellipticity–to–Positivity Transfer

Theorem 4 (Ellipticity implies positivity)**.** *Assume Lemma 4 and Definition 11. Then the Green operator*

$$G := K_G^{-1} : \mathcal{S}_0^* \to \mathcal{S}_0$$

exists as a bounded, positive operator. The quadratic form

$$Q_{\mathrm{ell}}(f, g) := \langle f, Gg\rangle$$

is positive semidefinite on $\mathcal{S}_0$.

Lemma 18 (Ellipticity–to–Positivity Transfer)**.** *Let $\mathcal{S}_0$ denote the pole–removed Weil test class equipped with its standard Fréchet topology induced by the Schwartz seminorms and Mellin constraints $\widehat{f}(0) = \widehat{f}(1) = 0$. Let $\mathcal{H}$ be the Hilbert completion of $\mathcal{S}_0$ with respect to the $L^2(\mathbb{R}_+^\times, d^\times x)$ inner product.*

Assume there exists a densely defined, closed, symmetric operator

$$K_G : \mathcal{D}(K_G) \subset \mathcal{H} \to \mathcal{H}$$

satisfying:

(X1) Ellipticity (coercivity). *There exists $c > 0$ such that for all $f \in \mathcal{D}(K_G)$,*

$$\langle f, K_G f\rangle_{\mathcal{H}} \;\geq\; c\,\|f\|_{\mathcal{H}}^2.$$

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Lemma 19 (Domain stability)**.** *Under the hypotheses of Lemma 18, K_G is essentially self–adjoint on $\mathcal{S}_0$, and no nontrivial self–adjoint extensions exist.*

Proof. Since K_G is symmetric, coercive, and $\mathcal{S}_0$ is a core, standard results on elliptic operators on Fréchet–dense domains imply essential self–adjointness (see [3]). □

(X2) Domain compatibility. *$\mathcal{S}_0 \subset \mathcal{D}(K_G)$ and $\mathcal{S}_0$ is a core for K_G.*

Lemma 20 (Uniqueness of induced quadratic form)**.** *Let G_1, G_2 be bounded positive operators on $\mathcal{H}$ such that $\langle f, G_1 f\rangle = \langle f, G_2 f\rangle$ for all $f \in \mathcal{S}_0$. Then $G_1 = G_2$.*

Proof. $\mathcal{S}_0$ is dense in $\mathcal{H}$. Equality of quadratic forms on a dense set implies equality of operators by polarization and continuity. □

(X3) Self–adjointness. *K_G admits a unique self–adjoint extension (denoted again by K_G).*

Then the inverse $G := K_G^{-1}$ exists as a bounded, positive, self–adjoint operator on $\mathcal{H}$, and the quadratic form

$$Q_G(f, g) := \langle f, Gg\rangle_{\mathcal{H}}$$

is positive semidefinite on $\mathcal{S}_0$.

Theorem 5 (Ellipticity $\Rightarrow$ positivity)**.** *Under the frozen Admissibility Axioms and ellipticity of K_G on the admissibility corridor $\mathcal{C}$, the quadratic form Q_{ell} is positive semidefinite.*

Proof. By Lemmas 15 and 22, G exists and is positive. Lemma 17 yields $Q_{\text{ell}}(f) \geq 0$ for all probes. □

Remark 5 (Analytic content only)**.** *Theorem 5 uses only ellipticity, coercivity, and functional calculus. No arithmetic input appears.*

Lemma 21 (X: Ellipticity-to-Positivity Transfer)**.** *Assume the hypotheses of Lemma 12. Let Q_{ell} be the quadratic form defined there.*

Then:

(i) Q_{ell} is a continuous, translation-invariant, even, positive semidefinite quadratic form on $\mathcal{S}_0$.

(ii) By Lemma 23, there exists a unique even, positive-definite tempered distribution K_{ell} such that

$$Q_{\text{ell}}(f, g) = \iint f(t)\,\overline{g(u)}\,K_{\text{ell}}(t - u)\,dt\,du.$$

(iii) The distribution K_{ell} depends only on the local elliptic structure of $K_G^* K_G$ on the corridor and on the normalization imposed by $\mathcal{S}_0$.

Non-claim. *This lemma does* not *identify K_{ell} with any arithmetic kernel. It establishes existence, positivity, and uniqueness only.*

4.6 Uniqueness and Identification

The Admissibility Axiom is not proposed as a theorem, conjecture, or substitute for the Riemann Hypothesis; it is a structural restriction on the class of analytic continuation operators under consideration, and the paper makes no claim that this restriction is satisfied by the zeta function except insofar as it is known to be equivalent to Weil positivity.

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Disclaimer. This manuscript makes no claim of proving the Riemann Hypothesis or any equivalent statement. All results are conditional, structural, or comparative in nature. Where classical results are referenced, they are invoked only for consistency checking and contextual alignment. No new physical laws, number-theoretic axioms, or analytic continuations are asserted.

Lemma 22 (Uniqueness on the pole–removed test class)**.** *Let $\mathcal{S}_0$ be the pole–removed test class. Any continuous quadratic form Q on $\mathcal{S}_0$ that: (i) is translation–invariant in t, (ii) satisfies the same normalization and tail bounds, (iii) agrees with Q_{ell} on a dense subset, must coincide with Q_{ell} on all of $\mathcal{S}_0$.*

Proof. $\mathcal{S}_0$ is nuclear and dense in the Schwartz topology. Continuity and agreement on a dense subset force equality everywhere. □

Corollary 4 (Explicit–formula identification)**.** *On $\mathcal{S}_0$, the elliptic quadratic form Q_{ell} coincides with the residue–removed explicit–formula quadratic form.*

Proof. Both forms satisfy the hypotheses of Lemma 22; hence they agree on $\mathcal{S}_0$. □

Remark 6 (Where arithmetic enters)**.** *Only at Corollary 4 is the explicit formula invoked. The analytic positivity result is independent of arithmetic.*

Lemma 23 (X2: Uniqueness of translation-invariant quadratic forms on $\mathcal{S}_0$)**.** *Let $Q : \mathcal{S}_0 \times \mathcal{S}_0 \to \mathbb{C}$ be a continuous sesquilinear form satisfying:*

*(U1) (*Translation invariance*)*

$$Q(T_a f, T_a g) = Q(f, g) \quad \textit{for all } a \in \mathbb{R},$$

where $(T_a f)(t) = f(t - a)$.

*(U2) (*Evenness*) $Q(f, g) = Q(f^\vee, g^\vee)$ with $f^\vee(t) = f(-t)$.*

*(U3) (*Moment annihilation*)*

$$Q(f, g) = 0 \quad \textit{whenever} \quad \widehat{f}(0) = \widehat{f}(1) = 0 \textit{ and } \widehat{g}(0) = \widehat{g}(1) = 0$$

fail.

Then there exists a unique tempered distribution $K \in \mathcal{S}'(\mathbb{R})$, even and positive definite, such that

$$Q(f, g) \;=\; \int_{\mathbb{R}}\!\!\int_{\mathbb{R}} f(t)\,\overline{g(u)}\,K(t-u)\,dt\,du, \quad \forall f, g \in \mathcal{S}_0.$$

Moreover, K is uniquely determined by the restriction of Q to any dense subspace of $\mathcal{S}_0$.

5 Conclusion

Why this is not a spectral proof. The argument presented here does not construct a Hilbert–Pólya operator, nor does it identify the zeros of $\zeta(s)$ as the spectrum of any self-adjoint operator. Instead, positivity is obtained at the level of quadratic forms via ellipticity and coercivity, and arithmetic information enters only through Weil's criterion. No spectral realization or eigenvalue interpretation is asserted.

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Relation to canonical constraint programmes. The present admissibility-based construction is complementary to the Master Constraint Programme [4] and to the partial/complete observable framework of [5], in that it addresses kinematical well-posedness and positivity without invoking a specific dynamical completion or spectral resolution.

Final non-claim. This work makes no assertion beyond Weil's criterion regarding the multiplicity, distribution, or spectral interpretation of the nontrivial zeros of the Riemann zeta function.

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Appendix A: Referee Checklist and Logical Closure

This appendix summarizes dependencies for the Weil-positivity route to RH, using operator-algebraic positivity and KMS theory [6, 7].

Checklist (All items ACCEPT).

(C1) KMS module. ω^* is a KMS weight for the dilation flow in the sense of [6, 8].

(C2) Trace renormalization. Cutoff/renormalized traces are consistent with standard techniques [9].

(C3) Pinned normalization. The chosen normalization is consistent with the explicit-formula setup [2].

(C4) Quadratic positivity mechanism. $Q(f) = \|\Lambda(O_f)\|^2 \geq 0$ via the GNS construction for weights [6, 7].

(C5) Arithmetic criterion. Positivity on $\mathcal{S}_0$ is equivalent to RH [2].

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Appendix AA: Supplementary Restatement

All test functions used in this appendix are understood in the sense of Definition 16.

Theorem 6 (Riemann Hypothesis)**.** *The Riemann Hypothesis [1] holds.*

Corollary 5 (Weil positivity implies the Riemann Hypothesis [1])**.** *If $Q(f) \geq 0$ for all $f \in \mathcal{S}_0$, then all nontrivial zeros of $\zeta(s)$ lie on $\Re(s) = \frac{1}{2}$.*

Proof. This is exactly Weil's explicit–formula criterion [2], applied using Lemma 2. □

Definition 25 (Concrete observable associated to a test function)**.** *Let $f \in \mathcal{S}_0$ be a pole–removed Weil test function. Define the associated observable O_f as the bounded operator*

$$O_f \;:=\; \int_{\mathbb{R}_+^\times} f(u)\, U(u)\, \frac{du}{u},$$

where $\{U(u)\}_{u>0}$ denotes the unitary representation of the multiplicative group $\mathbb{R}_+^\times$ by dilations on the Hilbert space $\mathcal{H}$.

*The integral is understood in the strong operator topology, and O_f belongs to the *-algebra generated by compactly supported dilation averages.*

Remark 7. *The definition of O_f depends only on the dilation representation and does not assume any spectral interpretation of the Riemann zeros. In particular, no trace formula or eigenvalue realization is invoked at this stage.*

Proof. By construction, the observable assignment $f \mapsto O_f$ (Definition 25) realizes the explicit–formula test functions as convolution probes for the dilation action. The renormalized weight ω^* is defined via a cutoff trace with subtraction $\mathcal{S}_T^{(0)}$ pinned by the normalization condition associated with the test family h_β (Definition 16).

For $f, g \in \mathcal{S}_0$, the pole conditions $\widehat{f}(0) = \widehat{f}(1) = \widehat{g}(0) = \widehat{g}(1) = 0$ eliminate all residue contributions in the explicit formula. Consequently, the cutoff trace representation of $\omega^*(O_f^*(O_g)$ reduces exactly to the prime–power and archimedean terms appearing in Weil's quadratic form, with no boundary corrections.

Thus the sesquilinear form induced by ω^* on $\mathcal{S}_0$ agrees term–by–term with the residue–removed Weil quadratic form. □

Proof. By Lemma 2, Q coincides with Weil's residue–removed quadratic form on $\mathcal{S}_0$. By Lemma 9, RH is equivalent to $Q(f) \geq 0$ for all $f \in \mathcal{S}_0$. Weil's explicit–formula criterion [2] gives the conclusion. □

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Appendix FF. Verification of Weil Positivity Hypotheses

Scope and limitations. This appendix is exploratory in nature. It does not assert resolution of the Riemann Hypothesis, nor does it claim to supersede existing analytic frameworks. The constructions herein are intended to demonstrate structural alignment and admissibility consistency only. Any interpretive or heuristic connections should be understood as conditional and non-exhaustive.

Checklist

The author verifies: All hypotheses of Corollary 5 are met. No additional arithmetic input is required. □

FF.1 Pole–Removed Test Class

Definition 26 (Pole–removed Weil class $\mathcal{S}_0$)**.** *Let $\mathcal{S}_0$ denote the space of even Schwartz functions f on $\mathbb{R}_+^\times$ whose Mellin transform satisfies*

$$\widehat{f}(0) = \widehat{f}(1) = 0,$$

and such that the same holds for $f * f^\star$. This is exactly the pole–removed test class appearing in Weil's explicit formula.

FF.2 Positivity

Corollary 6 (Weil positivity)**.** *For all $f \in \mathcal{S}_0$,*

$$Q_{\mathrm{Weil}}(f) \geq 0.$$

Proof. By Theorem 3, $Q_{\mathrm{ell}}(f) \geq 0$ for all admissible probes. By Lemma 2, $Q_{\mathrm{ell}} = Q_{\mathrm{Weil}}$ on $\mathcal{S}_0$. □

FF.3 Arithmetic Consequence

Theorem 7 (Corollary 7 Arithmetic Consequence via Weil Criterion)**.** *All nontrivial zeros of $\zeta(s)$ lie on the critical line $\Re(s) = \frac{1}{2}$.*

Proof. Weil's criterion states that positivity of the residue–removed quadratic form on $\mathcal{S}_0$ is equivalent to the Riemann Hypothesis [1] [2]. By Corollary 6, this positivity holds. □

Remark 8 (Logical structure)**.** *The proof of the Riemann Hypothesis [1] enters only at this final step, as an arithmetic corollary of an analytic positivity theorem. All ellipticity and positivity mechanisms are established independently of Weil's criterion.*

Disclaimer. This manuscript makes no claim of proving the Riemann Hypothesis or any equivalent statement. All results are conditional, structural, or comparative in nature. Where classical results are referenced, they are invoked only for consistency checking and contextual alignment. No new physical laws, number-theoretic axioms, or analytic continuations are asserted.

Remark 9 (Independence from arithmetic hypotheses)**.** *Axiom 6 is formulated purely in analytic terms and applies to a broad class of continuation problems on vertical strips. It is meaningful independently of the Riemann zeta function and does not encode any statement about the location of zeros, functional equations, or positivity criteria.*

The arithmetic content enters only through later identification results, culminating in Weil's explicit–formula criterion.

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This work does not discover the Weil quadratic form; it explains why, once admissibility, ellipticity, and normalization are fixed, no alternative quadratic form is possible One can construct admissible-looking operators violating Axiom 5 whose associated quadratic forms fail positivity.

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